

Decision Tree

1. Aim

To implement a **Decision Tree classification algorithm** to classify data into different categories by using a tree-like structure based on selected features and decision rules.

2. Problem Statement

Decision Tree is a supervised learning technique used to make decisions and classify data based on different features. The problem is to build a tree that divides the given dataset into suitable groups and predicts the correct class for new data by following a sequence of decision rules from the root node to a leaf node.

3. Solution

The problem is solved by constructing a **Decision Tree** from the given training data. At each step, the feature that provides the best separation of the data is selected as the decision node. The process continues until the data is classified into the required categories. The trained tree can then be used to predict the class of new data.

4. Algorithm

1. Start with the given training dataset.
2. Select the feature that provides the best split of the data.

3. Create a decision node using the selected feature.
4. Divide the dataset into smaller groups based on the feature values.
5. Repeat the splitting process for each group.
6. Stop splitting when the data is sufficiently classified or the stopping condition is reached.
7. Assign a class label to each leaf node.
8. Use the constructed tree to classify new data.
9. Display the decision tree and prediction result.

5. Program

```
from sklearn.tree import DecisionTreeClassifier,  
plot_tree  
import matplotlib.pyplot as plt
```

```
# Training data
```

```
X = [  
    [1, 1],  
    [1, 0],  
    [0, 1],  
    [0, 0],  
    [1, 1],  
    [0, 0]
```

```
]
```

```
# Target classes
```

```
y = [1, 1, 0, 0, 1, 0]
```

```
# Create Decision Tree
```

```
model = DecisionTreeClassifier(  
    criterion="entropy",  
    random_state=42  
)
```

```
# Train the model
```

```
model.fit(X, y)
```

```
# Predict a new sample
```

```
new_data = [[1, 1]]
```

```
prediction = model.predict(new_data)
```

```
print("Predicted Class:", prediction[0])
```

```
# Display Decision Tree
```

```
plt.figure(figsize=(8, 5))  
plot_tree(  
    model,  
    feature_names=["Feature 1", "Feature 2"],  
    class_names=["Class 0", "Class 1"],  
    filled=True  
)  
plt.show()
```

6. Result

The Decision Tree classification algorithm was successfully implemented using Python and Scikit-learn. The model learned decision rules from the given training data and successfully classified a new input based on its feature values.

Output:

Predicted Class: 1

Thus, the **Decision Tree successfully classified the given input data** by following the decision rules generated from the training dataset.

